

```
In [13]: import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.datasets import imdb
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras import models, layers
from sklearn.model_selection import train_test_split
import tensorflow as tf
```

```
In [15]: (X_train, y_train), (X_test, y_test) = imdb.load_data(num_words=10000)
```

```
In [17]: maxlen = 200

X_train = pad_sequences(X_train, maxlen=maxlen)
X_test = pad_sequences(X_test, maxlen=maxlen)
```

```
In [21]: from tensorflow.keras import models, layers

model = models.Sequential()

model.add(layers.Input(shape=(200,))) # sequence length

model.add(layers.Embedding(10000, 32))
model.add(layers.GlobalAveragePooling1D())

model.add(layers.Dense(50, activation="relu"))
model.add(layers.Dropout(0.3))
model.add(layers.Dense(1, activation="sigmoid"))

model.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape	Para
embedding_1 (Embedding)	(None, 200, 32)	320,
global_average_pooling1d_1 (GlobalAveragePooling1D)	(None, 32)	
dense_2 (Dense)	(None, 50)	1,
dropout_1 (Dropout)	(None, 50)	
dense_3 (Dense)	(None, 1)	

Total params: 321,701 (1.23 MB)

Trainable params: 321,701 (1.23 MB)

Non-trainable params: 0 (0.00 B)

```
In [23]: model.compile(
    optimizer="adam",
    loss="binary_crossentropy",
    metrics=["accuracy"]
)
```

```
results = model.fit(  
    X_train, y_train,  
    epochs=3,  
    batch_size=512,  
    validation_split=0.2  
)
```

Epoch 1/3

40/40 ————— 3s 30ms/step - accuracy: 0.6122 - loss: 0.6855 - val_acc
uracy: 0.6614 - val_loss: 0.6691

Epoch 2/3

40/40 ————— 1s 23ms/step - accuracy: 0.7347 - loss: 0.6357 - val_acc
uracy: 0.7618 - val_loss: 0.5842

Epoch 3/3

40/40 ————— 1s 32ms/step - accuracy: 0.7988 - loss: 0.5264 - val_acc
uracy: 0.8020 - val_loss: 0.4718

```
In [25]: score = model.evaluate(X_test, y_test)  
         print("Test accuracy:", score[1])
```

782/782 ————— 2s 3ms/step - accuracy: 0.7963 - loss: 0.4772
Test accuracy: 0.7963200211524963

In []: